

## Analog CMOS IC Design (EE651)

### Assignment- 4

Total Marks – 10 points

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Date: 05/09/2025

Submission Date: 12/09/2025 (11:55 PM)

**Note:** Below is the list of questions on **MOS Transistors** and **Small-Signal Parameters of a MOS Transistor**. You are required to:

- Perform the simulations using **Cadence Virtuoso** with **SCL 180 nm PDK**.
- Prepare a detailed report based on your work.
- Submit the report for evaluation.

#### Report Guidelines:

- Include a **Conclusion** section at the end, briefly summarizing your insights and observations from the design process.
  - Name the report file as: RollNumber\_Assignment4.pdf
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1. Design a Cascode Amplifier with active load the following configuration.

- Voltage Gain > 60 dB
- Output swing > 650 mV
- Power budget < 20  $\mu$ W (only for cascode circuit)
- Load capacitance = 1pF

Calculate the BW and Unity Gain Frequency of your designed amplifier. And write the gain margin and phase margin.

2. Compare with your **Assignment 3** CS with cascode amplifier by sweeping **CL** from **20 fF** to **1 pF** (use **parametric analysis**).
- For each CL, record **gain**, **BW**, and **GBW**.

Briefly explain the observed trends (e.g., pole shift with CL, impact on GBW and bandwidth).

Note: You have only one available power supply (VDD of 1.8V) and a reference current source of 1  $\mu$ A. The biasing needs to be done with the help of current mirrors and voltage divider circuits using MOS transistors (as in Assignment 2).

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Report should include,

- Transistor level circuit diagram (full system and individual sub-blocks) and sizing (with overdrive voltage) table of all transistors.
- Step-by-step design methodology/derivations/calculations (in brief).
- Overall power consumption and its block-wise breakup should be provided.
- Plots should be neatly labelled, and labels should be clearly visible
- Final results should be provided in tabular format
- Any other information as deemed necessary